VANESSA AUDREY

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EDUCATION

Master of Science, Mechanical Engineering, GPA 3.91/4.0; University of California, Irvine **09/2021 – 06/2023**

Bachelor of Science, Mechanical Engineering (Cum Laude); University of California, Irvine **09/2016 – 03/2019**

WORK EXPERIENCE

Mechanical Design Engineer | Progressive Suspension **01/2024 – 01/2025**

- Led full-cycle development of automotive suspension from concept to production in aftermarket industry
- Performed static FEA and thermal analysis on automotive components including structural supports
- Designed components with focus on manufacturability (DFM), cost reduction, and production efficiency
- Developed design validation plan (DVP) to validate prototypes and ensure product compliance per SAE standards
- Performed root cause analysis & corrective action for suspension damage and reduced overhead costs by 10%

Process Engineer Intern | Kulicke & Soffa Industries **01/2023 – 01/2024**

- Optimized ultrasonic welding through DOEs and statistics in semiconductors, increasing throughput by 5%
- Automated data analysis with Python, boosting workflow productivity by 20% and streamlining processes
- Used Minitab and Python for Cp/CpK and T-test to show visualizations and support process improvements
- Performed process characterization and root cause analysis to improve yield quality using CMM and profilometer

Mechanical Design Engineer Intern | ChargePoint, Inc. **06/2022 – 09/2022**

- Led end-to-end design initiatives to construct test fixture for evaluating EV charging performance
- Utilized Siemens NX to develop parametric models and assemblies for complex mechanical systems
- Managed part files, assemblies, and revisions in Teamcenter PLM to ensure accurate version control
- Demonstrated expertise in electro-mechanical components, i.e., electrical wiring and sensor interfaces
- Enhanced test fixture capabilities by installing pressure sensors, flow meters, power sources, PRTs

Research and Development Engineer | KYOCERA Precision Tools **08/2019 – 07/2021**

- Developed test procedures for conducting failure analysis and qualification tests on prototype tools
- Proficient in CNC milling, including G-code programming, setup, and precision machining for prototype parts
- Streamlined toolpaths and improved turnaround time by 40% by using Autodesk HSMWorks
- Composed an automated CMM measurement program, increasing workflow efficiency by 50%

MASTERS THESIS

Control-Oriented Kinematic Models for Grasping Device - https://escholarship.org/uc/item/145293c6

- Designed wearable assistive device in SolidWorks and validated grasping movement through MATLAB
- Utilized SolidWorks to design 3D-printed components, optimizing geometries to enhance structural integrity
- Published peer-reviewed papers focused on the modeling and prototyping of index-thumb finger system

CORE COMPETENCIES

CAD/CAM Software: Siemens NX, SolidWorks, Autodesk HSMWorks, CNC Milling

Simulation Software & Programming Languages: COMSOL FEA, MATLAB, Python, HTML

Testing & Analysis: CMM, Minitab, oscilloscopes, Keyence experimental equipment + sensors, strain gages

Project Management: Syspro ERP, Teamcenter, Confluence

PUBLICATIONS

1. "Robust Multi-Legged Walking Robots for Interactions with Different Terrains", ASME. J. Mechanisms Robotics. January 2024. https://doi.org/10.1115/1.4062303
2. "Natural Motion Grasping Design using Control-Oriented Kinematic Models", Proceedings of the ASME 2023 IDETC. 2023. https://doi.org/10.1115/DETC2023-116654
3. "Towards Creating Robust Walking Multi-Legged Robots for Traversing Different Terrains", Proceedings of the ASME 2022 IDETC August 2022. https://doi.org/10.1115/DETC2022-90614
4. "Towards Modeling Finger Joint Coordination for Natural Motion", Proceedings of the 2022 USCToMM Symposium on Mechanical Systems and Robotics. 2022. https://doi.org/10.1007/978-3-030-99826-4_6